Xin Yue Zhang

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Summary

MSc Systems & Control graduate with a mechanical engineering background, specialising in state estimation, sensor fusion and calibration of measurement systems. Experienced in Kalman filtering, closed-loop control and model-based calibration in Python and MATLAB, plus root cause analysis of live industrial software in C++, C# and MySQL.

Technical Skills

Programming: Python (NumPy, SciPy, pandas), MATLAB, SQL
Tools: Git, Docker, Azure DevOps, HeidiSQL, LaTeX
Estimation & Control: Kalman filter, Extended Kalman filter, maximum likelihood estimation, nonlinear least squares (Levenberg-Marquardt), sensor calibration, closed-loop control
Modelling & Sensing: Point spread function modelling, cubic spline interpolation, time-of-arrival localization, probabilistic sensor fusion, adaptive optics, Shack-Hartmann wavefront sensing
Languages: Dutch, English, Cantonese

Education

Delft University of Technology Sep 2023 - Apr 2026
MSc Systems and Control *Delft, NL*

Delft University of Technology Sep 2019 - Jul 2023
BSc Mechanical Engineering *Delft, NL*

University of British Columbia Jan 2022 - May 2022
Exchange student, Mechanical Engineering *Kelowna, CA*

Projects

Depth-dependent PSF Calibration for 3D Single-Molecule Localization Microscopy MSc Thesis
Python / MATLAB / Fiji

- Developed a novel calibration framework for 3D SMLM that accounts for depth-dependent optical aberrations.
- Wrote the Python code for a cubic spline model of the astigmatic point spread function, interpolated across depth.
- Reduced axial bias by up to 99% in simulation and by 80% on experimental bead data versus coverslip calibration.

Acoustic Localization of a Mobile Robot Using Microphone Arrays Course project
Sensor fusion

- Localized a mobile robot in 2D from time-of-arrival measurements across 8 microphones.
- Applied Kalman and Extended Kalman filters to improve trajectory estimation and quantify uncertainty.
- Preprocessed data and calibrated sensors to correct microphone bias and measurement noise.

Wavefront Correction Using Adaptive Optics Course project
Closed-loop control

- Implemented sensorless adaptive optics algorithms to correct static wavefront aberrations with a deformable mirror.
- Implemented wavefront sensing and reconstruction with a Shack-Hartmann sensor, enabling closed-loop aberration correction.

Work Experience

Junior Manufacturing Execution System Engineer Aug 2026 - Sep 2026
Batenburg Bellt *Zeist, NL*

- Traced the root cause of a live status failure across a multi-language production system (C++, C#, JavaScript, MySQL) and fixed it.
- Analysed how a legacy status dashboard got its data from old code and stored procedures, to reproduce the same data over MQTT.
- Used Git, Azure DevOps pull requests and Docker with C#, ASP.NET Core and MySQL in an industrial automation environment.

Host May 2024 - Jul 2026
Delft University of Technology *Delft, NL*

- Facilitated communication between visitors and nine internal departments.

Technical Specialist Jul 2022 - Sep 2023
Apple *Amsterdam, NL*

- Troubleshot and diagnosed technical issues with Apple hardware and software.
- Handled sensitive and confidential information in accordance with GDPR requirements.